Java Swing Unit Converter Assignment

https://gemini.google.com/app/825339c807205975

User prompt: what does my teacher want for this assignment

Response: Your teacher wants you to create a **Unit Converter GUI application** using Java Swing based on the following specifications:

Goal

Replicate/build a unit converter GUI application ("單位換算器").

Specifications & Layout Requirements

1. **Window Setup:**
    - **Size:** 480×280 pixels.
    - **Layout Manager:** `BorderLayout`.
2. **NORTH Region:**
    - Contains a `JComboBox` to select the conversion category:
        - **Length** (長度)
        - **Weight** (重量)
        - **Temperature** (溫度)
3. **CENTER Region:**
    - Contains **2 rows**.
    - Each row must have one `JTextField` and one `JComboBox` (representing the **Source Unit** on top and the **Target Unit** on the bottom).
4. **Dynamic Dropdown Logic:**
    - When the conversion category changes in the top `JComboBox`, the unit options in both inner `JComboBox` components must dynamically update to match:
        - **Length (長度):** Meter (公尺), Centimeter (公分), Inch (英吋), Foot (英呎)
        - **Weight (重量):** Kilogram (公斤), Gram (公克), Pound (磅), Ounce (盎司)
        - **Temperature (溫度):** Celsius (攝氏), Fahrenheit (華氏), Kelvin (克氏)
5. **SOUTH Region:**
    - Contains a **"Convert" (換算)** button.
    - Clicking this button takes the input number from the first `JTextField`, calculates the conversion based on the selected units, and displays the result in the second `JTextField`.

User prompt: create one for me

Response: Here is a complete, runnable Java Swing implementation that fulfills all of your teacher's requirements:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionListener;

public class UnitConverter extends JFrame {
    // UI Components
    private JComboBox<String> categoryBox;
    private JTextField inputField;
    private JComboBox<String> sourceUnitBox;
    private JTextField outputField;
    private JComboBox<String> targetUnitBox;
    private JButton convertButton;

    // Unit options for each category
    private final String[] lengthUnits = {"公尺", "公分", "英吋", "英呎"};
    private final String[] weightUnits = {"公斤", "公克", "磅", "盎司"};
    private final String[] tempUnits = {"攝氏", "華氏", "克氏"};

    public UnitConverter() {
        // Window Configuration: 480x280, BorderLayout
        setTitle("單位換算器");
        setSize(480, 280);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        setLayout(new BorderLayout(10, 10));

        // --- NORTH: Category Selection ---
        String[] categories = {"長度", "重量", "溫度"};
        categoryBox = new JComboBox<>(categories);
        JPanel northPanel = new JPanel();
        northPanel.add(new JLabel("換算類型："));
        northPanel.add(categoryBox);
        add(northPanel, BorderLayout.NORTH);

        // --- CENTER: Input/Output and Unit Dropdowns (2 Rows) ---
        JPanel centerPanel = new JPanel(new GridLayout(2, 2, 10, 10));
        centerPanel.setBorder(BorderFactory.createEmptyBorder(10, 20, 10, 20));

        inputField = new JTextField("1");
        sourceUnitBox = new JComboBox<>();
        outputField = new JTextField();
        outputField.setEditable(false); // Result display field
        targetUnitBox = new JComboBox<>();

        centerPanel.add(inputField);
        centerPanel.add(sourceUnitBox);
        centerPanel.add(outputField);
        centerPanel.add(targetUnitBox);

        add(centerPanel, BorderLayout.CENTER);

        // --- SOUTH: Convert Button ---
        convertButton = new JButton("換算");
        JPanel southPanel = new JPanel();
```

```
    southPanel.add(convertButton);
    add(southPanel, BorderLayout.SOUTH);

    // Populating units for the initial selection
    updateUnitBoxes();

    // --- Event Listeners ---
    // Dynamic unit update when category changes
    categoryBox.addActionListener(e -> updateUnitBoxes());

    // Perform calculation on button click
    convertButton.addActionListener(e -> performConversion());
}

// Dynamic Dropdown Logic
private void updateUnitBoxes() {
    sourceUnitBox.removeAllItems();
    targetUnitBox.removeAllItems();

    String selectedCategory = (String) categoryBox.getSelectedItem();
    String[] unitsToLoad;

    if ("重量".equals(selectedCategory)) {
        unitsToLoad = weightUnits;
    } else if ("溫度".equals(selectedCategory)) {
        unitsToLoad = tempUnits;
    } else {
        unitsToLoad = lengthUnits;
    }

    for (String unit : unitsToLoad) {
        sourceUnitBox.addItem(unit);
        targetUnitBox.addItem(unit);
    }

    if (targetUnitBox.getItemCount() > 1) {
        targetUnitBox.setSelectedIndex(1); // Default target to second option
    }
}

// Conversion Logic
private void performConversion() {
    try {
        double val = Double.parseDouble(inputField.getText().trim());
        String category = (String) categoryBox.getSelectedItem();
        String src = (String) sourceUnitBox.getSelectedItem();
        String tgt = (String) targetUnitBox.getSelectedItem();

        double result = 0;

        if ("長度".equals(category)) {
            result = convertLength(val, src, tgt);
        } else if ("重量".equals(category)) {
            result = convertWeight(val, src, tgt);
        } else if ("溫度".equals(category)) {
            result = convertTemperature(val, src, tgt);
        }
```

```
            outputField.setText(String.format("%.4f", result));
        } catch (NumberFormatException ex) {
            outputField.setText("請輸入有效數字");
        }
    }

    // Length Conversion (Base Unit: Meters)
    private double convertLength(double val, String src, String tgt) {
        double inMeters = switch (src) {
            case "公分" -> val / 100.0;
            case "英吋" -> val * 0.0254;
            case "英呎" -> val * 0.3048;
            default -> val; // 公尺
        };

        return switch (tgt) {
            case "公分" -> inMeters * 100.0;
            case "英吋" -> inMeters / 0.0254;
            case "英呎" -> inMeters / 0.3048;
            default -> inMeters; // 公尺
        };
    }

    // Weight Conversion (Base Unit: Kilograms)
    private double convertWeight(double val, String src, String tgt) {
        double inKg = switch (src) {
            case "公克" -> val / 1000.0;
            case "磅" -> val * 0.45359237;
            case "盎司" -> val * 0.0283495231;
            default -> val; // 公斤
        };

        return switch (tgt) {
            case "公克" -> inKg * 1000.0;
            case "磅" -> inKg / 0.45359237;
            case "盎司" -> inKg / 0.0283495231;
            default -> inKg; // 公斤
        };
    }

    // Temperature Conversion
    private double convertTemperature(double val, String src, String tgt) {
        if (src.equals(tgt)) return val;

        // Convert source to Celsius first
        double inCelsius = switch (src) {
            case "華氏" -> (val - 32) * 5.0 / 9.0;
            case "克氏" -> val - 273.15;
            default -> val; // 攝氏
        };

        // Convert Celsius to target
```

```
        return switch (tgt) {
            case "華氏" -> (inCelsius * 9.0 / 5.0) + 32;
            case "克氏" -> inCelsius + 273.15;
            default -> inCelsius; // 攝氏
        };
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            new UnitConverter().setVisible(true);
        });
    }
}
```

Would you like me to add explanations for how any specific part of this code works?

User prompt: please make sure the code that you're using is aligned with what i have learned in class

Response: Here is the refactored code aligned specifically with your course presentation styles and patterns:

- **Extends `JFrame` & Implements `ActionListener`:** Follows the classic delegation event model style shown in slides 28–30 (e.g., `Ch17_7`).
- **Layouts:** Uses `BorderLayout`, `FlowLayout`, and `GridLayout` without unnecessary lambda expressions or modern switch syntax.
- **Explicit Listener Registration:** Registers listeners using `addActionListener(this)` and routes actions inside `actionPerformed(ActionEvent e)`.

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class UnitConverter extends JFrame implements ActionListener {
    // Components
    private JComboBox categoryBox;
    private JTextField inputField;
    private JComboBox sourceUnitBox;
    private JTextField outputField;
    private JComboBox targetUnitBox;
    private JButton convertButton;

    // Unit Arrays
    private String[] lengthUnits = {"公尺", "公分", "英吋", "英呎"};
    private String[] weightUnits = {"公斤", "公克", "磅", "盎司"};
    private String[] tempUnits = {"攝氏", "華氏", "克氏"};

    public UnitConverter() {
        setTitle("單位換算器");
        setSize(480, 280);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        // Window main layout
        setLayout(new BorderLayout());
```

```
    // --- NORTH: Category Selection ---
    String[] categories = {"長度", "重量", "溫度"};
    categoryBox = new JComboBox(categories);
    categoryBox.addActionListener(this);

    JPanel northPanel = new JPanel(new FlowLayout());
    northPanel.add(new JLabel("換算類型："));
    northPanel.add(categoryBox);
    add(northPanel, BorderLayout.NORTH);

    // --- CENTER: 2 Rows (TextField + JComboBox) ---
    JPanel centerPanel = new JPanel(new GridLayout(2, 2, 10, 10));

    inputField = new JTextField("1");
    sourceUnitBox = new JComboBox();
    outputField = new JTextField();
    outputField.setEditable(false);
    targetUnitBox = new JComboBox();

    centerPanel.add(inputField);
    centerPanel.add(sourceUnitBox);
    centerPanel.add(outputField);
    centerPanel.add(targetUnitBox);

    add(centerPanel, BorderLayout.CENTER);

    // --- SOUTH: Convert Button ---
    convertButton = new JButton("換算");
    convertButton.addActionListener(this);

    JPanel southPanel = new JPanel(new FlowLayout());
    southPanel.add(convertButton);
    add(southPanel, BorderLayout.SOUTH);

    // Initial setup for default category
    updateUnitBoxes();
}

// Dynamic dropdown updates based on category
private void updateUnitBoxes() {
    sourceUnitBox.removeAllItems();
    targetUnitBox.removeAllItems();

    String selectedCategory = (String) categoryBox.getSelectedItem();
    String[] unitsToLoad;

    if (selectedCategory.equals("重量")) {
        unitsToLoad = weightUnits;
    } else if (selectedCategory.equals("溫度")) {
        unitsToLoad = tempUnits;
    } else {
        unitsToLoad = lengthUnits;
    }

    for (int i = 0; i < unitsToLoad.length; i++) {
        sourceUnitBox.addItem(unitsToLoad[i]);
        targetUnitBox.addItem(unitsToLoad[i]);
```

```
    }

    if (targetUnitBox.getItemCount() > 1) {
        targetUnitBox.setSelectedIndex(1);
    }
}

// Action listener event handler
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == categoryBox) {
        updateUnitBoxes();
    } else if (e.getSource() == convertButton) {
        performConversion();
    }
}

// Conversion logic
private void performConversion() {
    try {
        double val = Double.parseDouble(inputField.getText().trim());
        String category = (String) categoryBox.getSelectedItem();
        String src = (String) sourceUnitBox.getSelectedItem();
        String tgt = (String) targetUnitBox.getSelectedItem();

        double result = 0;

        if (category.equals("長度")) {
            result = convertLength(val, src, tgt);
        } else if (category.equals("重量")) {
            result = convertWeight(val, src, tgt);
        } else if (category.equals("溫度")) {
            result = convertTemperature(val, src, tgt);
        }

        outputField.setText(String.format("%.4f", result));
    } catch (NumberFormatException ex) {
        outputField.setText("請輸入有效數字");
    }
}

// Length conversion (Base unit: meters)
private double convertLength(double val, String src, String tgt) {
    double inMeters = val;
    if (src.equals("公分")) {
        inMeters = val / 100.0;
    } else if (src.equals("英吋")) {
        inMeters = val * 0.0254;
    } else if (src.equals("英呎")) {
        inMeters = val * 0.3048;
    }

    if (tgt.equals("公分")) {
        return inMeters * 100.0;
    } else if (tgt.equals("英吋")) {
        return inMeters / 0.0254;
    } else if (tgt.equals("英呎")) {
```

```
            return inMeters / 0.3048;
        }
        return inMeters;
    }

    // Weight conversion (Base unit: kilograms)
    private double convertWeight(double val, String src, String tgt) {
        double inKg = val;
        if (src.equals("公克")) {
            inKg = val / 1000.0;
        } else if (src.equals("磅")) {
            inKg = val * 0.45359237;
        } else if (src.equals("盎司")) {
            inKg = val * 0.0283495231;
        }

        if (tgt.equals("公克")) {
            return inKg * 1000.0;
        } else if (tgt.equals("磅")) {
            return inKg / 0.45359237;
        } else if (tgt.equals("盎司")) {
            return inKg / 0.0283495231;
        }
        return inKg;
    }

    // Temperature conversion
    private double convertTemperature(double val, String src, String tgt) {
        if (src.equals(tgt)) return val;

        double inCelsius = val;
        if (src.equals("華氏")) {
            inCelsius = (val - 32) * 5.0 / 9.0;
        } else if (src.equals("克氏")) {
            inCelsius = val - 273.15;
        }

        if (tgt.equals("華氏")) {
            return (inCelsius * 9.0 / 5.0) + 32;
        } else if (tgt.equals("克氏")) {
            return inCelsius + 273.15;
        }
        return inCelsius;
    }

    public static void main(String[] args) {
        UnitConverter frame = new UnitConverter();
        frame.setVisible(true);
    }
}
```